

Robin Lange

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Robotics and Autonomous Systems | Perception, Mission Logic and Physical Testing

Education

Georgia Southern University

Aug. 2023–May 2027

B.S. Mechanical Engineering, Vehicle Design & Testing concentration

GPA: 4.00/4.00

President's List; Honors College Scholar; Tau Beta Pi (junior upper eighth); Allen E. Paulson Scholar.

Descartes-Gymnasium Neuburg an der Donau

Sept. 2014–June 2023

Allgemeine Hochschulreife / General Higher Education Entrance Qualification

Germany

Honors Scholar; top 10% of graduating class. Final German grade: 1.2 (1.0 highest); 791/900 points.

Research Experience

Institute for Human & Machine Cognition (IHMC)

May–Aug. 2026

National Center for Collaborative Autonomy (NCCA)

Collaborative Robotics Autonomy Intern

Pensacola, FL

- Built a Rust/Python collaborative-autonomy codebase for two Boston Dynamics Spot robots; implemented pickup/drop-off sequencing and Joint Activity Graph coordination for group decisions.
- Combined inverse kinematics for coarse approach with local perception for final grasping, enabling hardware pickups despite tracking drift and imperfect perception.
- Implemented coordinate transforms and topic-based Zenoh communication so agents with different capabilities could share task state; a successor later added a drone.
- Integrated Motive motion capture and MCAP/Foxglove logging; instrumented a manually operated Husky AGV as a mobile, raised drop-off platform.

Intelligent Vehicles Navigation Research Laboratory

Oct. 2023–present

Georgia Southern University

Undergraduate Researcher

Statesboro, GA

- Integrated visual SLAM, nvblox mapping, Nav2 and behavior-tree mission logic on Jetson/RealSense UAV hardware; evaluated localization in a 110-second cart trial (14.5 cm RMSE on valid reference segments).
- Developed a LiDAR hand-motion interface for UAV control; coauthored two published SAE papers on lane perception and vehicle-simulation calibration.
- Honors thesis: commissioning a PD longitudinal-control benchmark in MATLAB and AVL; planning matched physical, always-on learned and reliability-gated preview comparisons under the same safety-supervised MPC.

CARISSMA Institute of Safety in Future Mobility (C-ISAFE)

July 1–Aug. 15, 2025

Technische Hochschule Ingolstadt

Vehicle Testing Research Intern

Ingolstadt, Germany

- Investigated how a soft truck-rear target appeared to radar, LiDAR and camera sensors; supported vehicle measurements and analyzed qualitative LiDAR/4D-radar returns with team scripts.
- Used the sensor observations to refine the target's tarpaulin covering and geometry; built its quick-assembly and fold-away mechanisms from an earlier CAD/foam concept.

Fraunhofer Institute for Transportation and Infrastructure Systems IVI

June 1–Aug. 11, 2024

Advanced Air Mobility research

Autonomous Systems and AI Intern

Ingolstadt, Germany

- Developed “Mission AI” decision and execution logic in Python, ROS 2 and py_trees; compared state-machine and behavior-tree approaches within an existing autonomy framework.
- Tested flight-phase transitions and emergency interruption in simulation, evaluated resource use and documented results for the research team.

Technical Skills

Programming: Python, Rust, MATLAB, C++; functional/procedural programming; entity-component-system (ECS) patterns.

Robotics: ROS 2/DDS, Zenoh, Protocol Buffers, Nav2; inverse kinematics, behavior trees, coordinate transforms, visual SLAM.

Hardware and tools: Boston Dynamics Spot, TurtleBot 4/Create 3, motion capture, LiDAR, RGB-D cameras, Jetson, Gazebo, SolidWorks, Git/Linux, MCAP/Foxglove.

Selected Awards and Scholarships

Third Place Undergraduate, Gulfstream Innovation, Engineering and Flight Excellence Awards – April 24, 2026. *Development of a Pilot AI for Autonomous UAV Inspection in Industry 5.0 Using Visual SLAM and Behavior Trees*; CEC Research Symposium.

Second Place Undergraduate, Georgia Power Innovation Awards – April 25, 2025. *Drone Guiding Through Equivalent Hand Movement Transversion*; CEC Research Symposium.

NCUR Engineering Division Video Contest Awardee – Council on Undergraduate Research, 2026.

Honors College Continuing Scholarship (2024–27; fewer than 15% of Honors College students receive college scholarships); Charles Robert Pound, Sr. Scholarship (2026–27); Paulson Scholar (2025); Vickie Hawkins Scholarship (2024).

Published Research

Soloiu, V., Sutton, T., Mehrzed, S., **Lange, R.**, Zimmerman, C., and Peralta Lopez, G. “Data-Driven Calibration of Vehicle Simulations for Accurate Control Response in Development and Testing.” *SAE Technical Paper 2026-01-0058*, 2026. doi:10.4271/2026-01-0058.

Soloiu, V., Mehrzed, S., Kroeger, L., Pierce, K., Sutton, T., and **Lange, R.** “Development of a Rule-Based Lidar Lane-Keeping Assistance System for Autonomous Vehicles Applications.” *SAE Technical Paper 2025-01-8025*, 2025. doi:10.4271/2025-01-8025.

Preprint and Manuscript in Preparation

Soloiu, V., Sutton, T., **Lange, R.**, and Zimmerman, C. “Warehouse Platooning Framework Using a Sensor-Rich Leader and Minimal Perception Follower.” SSRN 7067358. Preprint; not peer reviewed.

Lange, R., Sezgin, F., Inderst, M., Sutton, T., Lugner, R., Brandmeier, T., and Soloiu, V. “Design and Validation of a Soft Crash Target for Multimodal Sensor-Based Truck Underride Testing.” Manuscript in preparation.

Lange, R. Honors thesis in progress: reliability-gated lead-vehicle prediction for safety-supervised model predictive ACC. PD benchmark commissioning is underway; matched physical and learned-preview comparisons are planned.

Additional Research Presentations

Southern Regional Honors Council Conference, 2026, Montgomery, AL: Intelligent UAV Navigation with V-SLAM and Behavior Trees for Industry 5.0 Applications.

Southern Regional Honors Council Conference, March 2025, Chattanooga, TN: 3D-Space Hand Trajectory Tracking for Intuitive UAV Control.

Teaching and Academic Preparation

Physics I/II teaching assistant (Jan. 2025–present): evaluate experimental reports.

Computing for Engineers laboratory assistant (Jan.–May 2024): guided MATLAB and programming labs.

First-Year Experience peer leader (Aug.–Dec. 2024).

Selected completed coursework: Calculus I–III, Differential Equations, Numerical Methods, Dynamics, Mechatronics, Circuit Analysis, Program Design, Logic Circuit Design, Measurement & Instrumentation, Vehicle Design & Testing, Mechanism Design, Machine Design.

Languages: German (native), English (fluent), French (DELF B1).

Activity History Complementing Education on Page 1

Georgia Southern University

Aug. 2023–May 2027

B.S. Mechanical Engineering, Vehicle Design & Testing concentration

Statesboro, GA

Full-time undergraduate study with concurrent research, teaching and the summer internships listed on pages 1–2.

Wittelsbacher Golfclub

June–Aug. 2023

Administrative and Customer Service Assistant

Neuburg an der Donau, Germany

Additional Employment and Teaching

Physics Teaching Assistant

Jan. 2025–present

Principles of Physics I and II, Georgia Southern University

First-Year Experience Peer Leader

Aug.–Dec. 2024

Georgia Southern University

Laboratory Assistant, Computing for Engineers

Jan.–May 2024

Georgia Southern University

Event Assistant

Dec. 2023

Maritim Hotel & Congress Centrum

Ingolstadt, Germany

Student Assistant

Dec. 2023

Eagle Dining Services

Statesboro, GA

Tutor

Jan.–July 2023

Genius Tutoring and Languages

Neuburg an der Donau, Germany

Mathematics, physics, chemistry, information technology and English.

School work-experience placement

Apr. 19–20, 2022

Airbus Helicopters

Donauwörth, Germany

Exposure to the aircraft-mechanic field.

Short aviation/technical placement

June 7–10, 2022

Airbus Defence and Space

Manching, Germany

Observed aircraft assembly and manufactured model prototypes using workshop equipment.

Earlier Academic Work

Physics W-Seminar paper: “Der Spin des Golfballs.” Descartes-Gymnasium Neuburg an der Donau.